

Gyuwon Na

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EDUCATION

DigiPen Institute of Technology Redmond, WA	Aug 2026 – Expected May 2028
B.S. Computer Science in Real-Time Interactive Simulation GPA: 4.0 / 4.0	
Keimyung University Daegu, South Korea	Mar 2023 - Jun 2026
KMU-DigiPen Dual-Degree Program GPA: 4.38 / 4.5	
Coursework: Computer Graphics, Data Structures, Computer Networks, Digital Signal Processing, Circuit Analysis	

TECHNICAL SKILLS

Languages: C++, Python, C, GLSL
Tools & Platforms: Linux, CMake, Git/GitHub, ROS 2 / ROS 1, Docker, MATLAB
Libraries & Engines: OpenGL, WebGL, OpenCV, PyTorch, CUDA, Unity

RESEARCH EXPERIENCE

ISIP Lab Undergraduate Researcher Daegu, South Korea	Aug 2025 - Present
- Designed a 1.43M-parameter, five-stage geometric alignment framework using Clifford-inspired scalar, vector, and rotor-related features with transformer, residual, and energy-based refinement for severe in-plane rotations. - Reduced mean rotation error on the 100 hardest cases from 40.49° to 17.01° (58%); built COCO, HPatches, and MegaDepth evaluations with baseline, ablation, and failure-case analysis. - First author of an Image and Vision Computing manuscript under review - First author of a KMMS Fall Conference poster on adaptive binarization-based lane detection.	

PROJECTS

SEA:ME Hackathon - Vision-Only Driving System Lead Autonomy Engineer	Jul 2026
- Led end-to-end integration of a single-camera ROS 2 autonomy stack spanning lane perception, NCNN detection, mission state machines, Pure Pursuit control, debugging tools, and embedded actuator interfaces. - Increased end-to-end perception from 6 FPS to a stable 20 FPS (3.3x) through shared-memory process isolation, mission-specific ROIs, fresh-frame queues, and ROS 2 execution tuning. - Completed the course in 56 seconds, finishing only 0.03 seconds behind 3rd place.	
KAIST Mobility Challenge - V2X Cooperative Driving Systems Engineer	Oct 2025 - Feb 2026
- Built V2X cooperative decision and control logic for four autonomous vehicles negotiating roundabouts and four-way intersections, including stop-or-go collision avoidance from shared pose and velocity state. - Implemented MPC-inspired Pure Pursuit tracking, sim-to-real braking compensation, and YAML-configured per-vehicle tuning.	
MANZO - Custom C++ / OpenGL Game Engine Technical Lead	Jun 2024 - Jun 2025
- Led a year-long engine and game project, implementing vector-font rendering, underwater visibility shaders, reusable UI systems, positional audio, and rendering diagnostics with RenderDoc. - Built autonomous fish behaviors with pathfinding, terrain avoidance, flocking, and quadtree-based spatial partitioning to reduce per-frame spatial queries.	
LEGO Finder - Prompt-Guided Visual Search Computer Vision Project	Apr 2026 - Jun 2026
Implemented a real-time, prompt-guided LEGO finder that narrows search with ROI masking, queries an offline HNSW index, verifies matches with RANSAC, and highlights targets in AR.	

AWARDS

SEA:ME Hackathon Silver Award, 4 th Overall	Jul 2026
KAIST Mobility Challenge Excellence Award, 3 rd Overall	Feb 2026
2025 HL Future Mobility Award Encouragement Award	Sep 2025
2025 Micro Degree Expo Outstanding Award, 2 nd Place	Jun 2025
Artificial Intelligence Software Competition – C Programming Division Excellence Award, 3 rd Place	Aug 2024

LEADERSHIP EXPERIENCE

Keimyung University Class Representative & Student Council Member	Sep 2023 - Jun 2025
- Represented students as the primary liaison with faculty, consolidating academic concerns and communicating decisions across the dual-degree program. - Coordinated academic events, student activities, and departmental programs while aligning schedules and responsibilities across students, faculty, and organizers.	